What is the difference between CMD & ENTRYPOINT

1 - CMD

ده بيحددلي default command الي بيشتغل لما ال container يبدا
يعني اول ما اعمل run لل container بتاعي هيشتتغل علطول بس لو انا باصيت ليه argument هيتجاهل default command ويشغل ال argument

2 - ENTRYPOINT

هو عكس ال cmd مش بيتلغي لو باصيت ليه argument بينفذها مع ال command بتاعه

What is the difference between COPY & ADD

1 - COPY

ده بينسخ الملفات والفولدرات من الجهاز بتاعي لل image

2 -Add

هو تقريبا زي copy بس زياده عليه انه ممكن يفك ضغط الملفات ويحطها جوا ال image وكمان لو اديته url هينزل الملفات بتاعته ويحطها جوا ال image

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- Run the container hello-world

```
~> docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/
```

- Check the container status

```
~> docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
```

- Start the stopped container

```
~> docker ps -a
CONTAINER ID   IMAGE         COMMAND    CREATED          STATUS
ORTS     NAMES
53400d2445c3   hello-world   "/hello"   4 minutes ago    Exited (0) 4 minutes ago
         amazing_burnell
d15143c574ec   hello-world   "/hello"   11 minutes ago   Exited (0) 11 minutes ago
         xenodochial_elion
f048ffa30bf6   hello-world   "/hello"   24 hours ago     Exited (0) 24 hours ago
         epic_chatterjee
b7b428f70f40   hello-world   "/hello"   24 hours ago     Exited (0) 24 hours ago
         affectionate_hawking
~> docker start 53400d2445c3
53400d2445c3
```

- Remove the container

```
~> docker rm 53400d2445c3
53400d2445c3
```

- Remove the image

```
~> docker rmi 1b44b5a3e06a
Error response from daemon: conflict: unable to delete 1b44b5a3e06a (must be forced) - image is being
used by stopped container b7b428f70f40
exit 1
~> docker rmi -f 1b44b5a3e06a
Untagged: hello-world:latest
Untagged: hello-world@sha256:ef54e839ef541993b4e87f25e752f7cf4238fa55f017957c2eb44077083d7a6a
Deleted: sha256:1b44b5a3e06a9aae883e7bf25e45c100be0bb81a0e01b32de604f3ac44711634
~> |
```

- Run container centos or ubuntu in an interactive mode

```
~> docker run -it ubuntu bash
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
01d7766a2e4a: Pull complete
Digest: sha256:d1e2e92c075e5ca139d51a140fff46f84315c0fdce203eab2807c7e495eff4f9
Status: Downloaded newer image for ubuntu:latest
root@b8770380b6ea:/# ls
bin   dev  home  lib64  mnt  proc  run   srv  tmp  var
boot  etc  lib   media  opt  root  sbin  sys  usr
```

- Run the following command in the container “echo docker ”

```
root@b8770380b6ea:/# echo docker
docker
```

- Open a bash shell in the container and touch a file named hello-docker

```
root@b8770380b6ea:/# touch hello-docker
root@b8770380b6ea:/# ls
bin   dev  hello-docker  lib    media  opt   root  sbin  sys  usr
boot  etc  home          lib64  mnt    proc  run   srv   tmp  var
```

Stop the container and remove it. Write your comment about the file hello-docker

الملف هيتمسح

```
~> docker ps -a
CONTAINER ID   IMAGE          COMMAND    CREATED
b8770380b6ea   ubuntu         "bash"     6 minutes ago
are
f048ffa30bf6   1b44b5a3e06a   "/hello"   24 hours ago
atterjee
b7b428f70f40   1b44b5a3e06a   "/hello"   24 hours ago
onate_hawking
~> docker rm b8770380b6ea
b8770380b6ea
~>
```

- Remove all stopped containers

```
~> docker container prune
WARNING! This will remove all stopped containers.
Are you sure you want to continue? [y/N] y
Deleted Containers:
f048ffa30bf62fbfc2f20582ffba1f9c00b96d45173c35d33d916803f6a77156
b7b428f70f405298ab17b1305e77c85785b747558d8503ef3e580fdf8efd9722

Total reclaimed space: 0B
~>
```

Deploy a MySQL database called app-database. Use the mysql latest image, and use the -e flag to set MYSQL_ROOT_PASSWORD to P4sSw0rd0!. The container should run in the background.

```
~> docker run -d --name app-database -e MYSQL_ROOT_PASSWORD=yazz mysql:latest
Unable to find image 'mysql:latest' locally
latest: Pulling from library/mysql
74a8e4bbd9fe: Pull complete
357926362d31: Pull complete
f3702ac323ca: Pull complete
658630c937c6: Pull complete
ab1b8ea72826: Pull complete
61941d436e88: Pull complete
c89a19cdad3b: Pull complete
00af5f73db53: Pull complete
8314ff001dec: Pull complete
4a629f1008ff: Pull complete
Digest: sha256:e5dc14f6e01c3e577e669337d2855c6d1561b30d8ef2c592e63e4e8a9a52650a
Status: Downloaded newer image for mysql:latest
ad9fdb3d9876c440f1b7c60b86c72fd3097a5c0d9c7417cb93fce97f7a39a842
~>
```

- Run the image Nginx

```
~> docker run -d --name nginx-server -p 8080:80 nginx
c3c5fd4dfbb60e586b2442910aef53ca39ed07ec263b3b39daf7939bf7dd8998
```

- Add html static files to the container and make sure they are accessible

```
~> docker exec -it nginx-server bash
root@c3c5fd4dfbb6:/# cd /usr/share/nginx/html
root@c3c5fd4dfbb6:/usr/share/nginx/html# ls
50x.html  index.html
root@c3c5fd4dfbb6:/usr/share/nginx/html# echo "<h1>Hello from yasser tawfik</h1>"
 > index.html
root@c3c5fd4dfbb6:/usr/share/nginx/html#
```

- Commit the container with image name IMAGE_NAME

```
~> docker commit nginx-server my-nginx-image
sha256:0ae3f6dbd40c5781325df176060c83c05b7575d1f90a2fdbb5b673dab36bab98
```

- Create a python simple app

```
~> mkdir python-app
~> cd python-app
~/python-app> nano app.py
~/python-app>
```

- Create a dockerfile to containerize the python app

```
  GNU nano 7.2                    Dockerfile
FROM python:3.11-slim

WORKDIR /app

COPY app.py /app/

CMD ["python","app.py"]
```

- Build the image and test it

```
~/python-app> docker build -t python-app .
DEPRECATED: The legacy builder is deprecated and will be removed in a future rele
ase.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  3.072kB
Step 1/4 : FROM python:3.11-slim
3.11-slim: Pulling from library/python
206356c42440: Already exists
206709444425: Pulling fs layer
bbf418262cbe: Pulling fs layer
43272e9c9d4b: Pulling fs layer
43272e9c9d4b: Verifying Checksum
43272e9c9d4b: Download complete
206709444425: Verifying Checksum
206709444425: Download complete
206709444425: Pull complete
bbf418262cbe: Verifying Checksum
bbf418262cbe: Download complete
bbf418262cbe: Pull complete
43272e9c9d4b: Pull complete
Digest: sha256:d6e4d224f70f9e0172a06a3a2eba2f768eb146811a349278b38fff3a36463b47
Status: Downloaded newer image for python:3.11-slim
 ---> 7c68b5683872
Step 2/4 : WORKDIR /app
 ---> Running in c803a108f81d
 ---> Removed intermediate container c803a108f81d
 ---> b1e50bc99e60
Step 3/4 : COPY app.py /app/
 ---> 9036a3fb4fce
Step 4/4 : CMD ["python","app.py"]
 ---> Running in 061a84b76552
 ---> Removed intermediate container 061a84b76552
 ---> c6f12553e924
Successfully built c6f12553e924
Successfully tagged python-app:latest
```

```
~/python-app> docker run --name python-container  python-app
hallo from yasser
~/python-app>
```

- Push the created image into your docker hub repo

```
Login Succeeded
~/python-app> docker tag python-app yassertawfik48/python-app:latest

~/python-app> docker push yassertawfik48/python-app:latest
```

```
~> docker images
REPOSITORY                  TAG         IMAGE ID       CREATED          SIZE
python-app                  latest      c6f12553e924   14 minutes ago   124MB
yassertawfik48/python-app   latest      c6f12553e924   14 minutes ago   124MB
yassertito682/python-app    latest      c6f12553e924   14 minutes ago   124MB
my-nginx-image              latest      0ae3f6dbd40c   23 minutes ago   161MB
python                      3.11-slim   7c68b5683872   2 days ago       124MB
nginx                       latest      fd204fe2f750   9 days ago       161MB
mysql                       latest      8458fdfc578c   2 weeks ago      922MB
ubuntu                      latest      bbdabce66f1b   3 weeks ago      78.1MB
~>
```